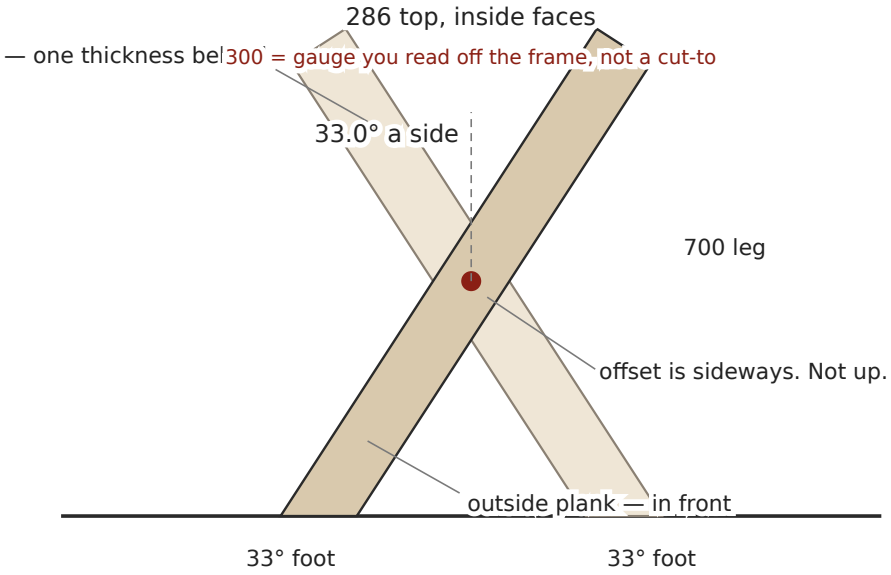
X-Leg Herb Trough

plan · v9

- Two X-leg frames carry a **900** trough.
- Legs **4×700**. Cross planks **8×900**. Planks **80 wide, 23 thick**.
- **Gauge 300**, inside corner to inside corner.
- Read the gauge off the frame. Never cut to it.
- Legs do not notch. One plank in front, one behind, **one thickness apart**.
- The ends are **scribed** off an offcut. Not a number.
- One trim only: **one plank thickness** off the second floor plank.
- Shown at 900 long, 4 planks deep. All in mm.

X JOINT - stage 1



The joint decides the box.

- Two 700 legs, crossed. **Both feet reach the floor.**
- The inside plank sits one thickness behind the outside plank.
- That offset is **sideways**, out of the page. Nothing rides up.
- Two frames. **Mirror pair:** the outside plank swaps sides.
- 35 mm part-threaded: through the near plank, bites the far one.
- Span between the inner faces: **286**. Worked out, not measured.
- **2 × (350·sin33° − 40÷cos33°) = 286**
- **300** is the gauge on the standing frame. Not a cut-to.
- Foot flush on the clamped plank, top at 300. Frame stands true. Both frames match.

CUT LIST

PART	QTY	LENGTH	WIDTH	THICK	NOTE
Leg	4	700	80	23	one end cut 33°, all four one setting
Cross plank (wall)	8	900	80	23	4 front, 4 back
Side piece	6	scribed	80	23	3 per end, marked off an offcut
- of those, the bottom pair	2	900	80	23	first plank's inner long edge cut 22.5° into the V; second trimmed one plank thickness

BUY IT IN YOUR YARD'S STOCK

- **Timber: 6 lengths of 2 m**, 80 wide × 23 thick.
- Legs: 4×700 = 2,800. **2 lengths.** Two legs per length.
- Walls: 8×900 = 7,200. **4 lengths.** Two per length, 200 off each.
- Offcuts left: two 600s, four 200s.
- **Watch the sides.** Six side pieces come off the two 600s. Longest first.
- Needs more than 600? Buy a 7th length.
- **The trim is 23.** One plank thickness. Your number, same rule.
- **Buying 90 wide instead?** Two numbers change. Nothing else does.
- Crossing mark: **45 across**. Half of 90, not 40.
- Top span between leg inner faces: **274**. Same formula, 45 in place of 40.
- 700, 900, 33°, 22.5°, the thickness trim: all the same.
- The drawing here is at 80 wide.
- Untreated pine or hardwood. Untreated: it is a food bed.
- **35 mm part-threaded screws** for the crossing and the side courses.
- A **6 mm or 10 mm bolt** through each crossing if you want it to never move.
- Stainless or coated only. Wood oil (Woodoc) outside and on the feet.

The rules read this first

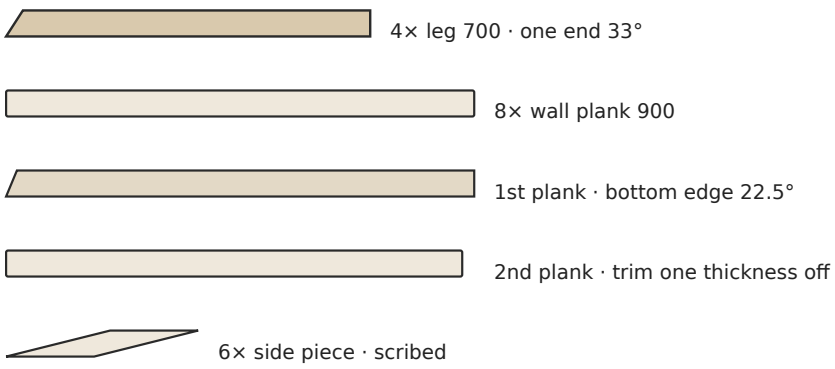
Two rules that decide this build

- **One saw setting for all four feet.** Cut every 33° end off the same stop. One foot cut on its own and the box stands on three legs.
- **Lock the crossing.** A single screw lets the X scissor. Add 3-4 screws, or bolt it.

Locked in from your build photos (14 Oct)

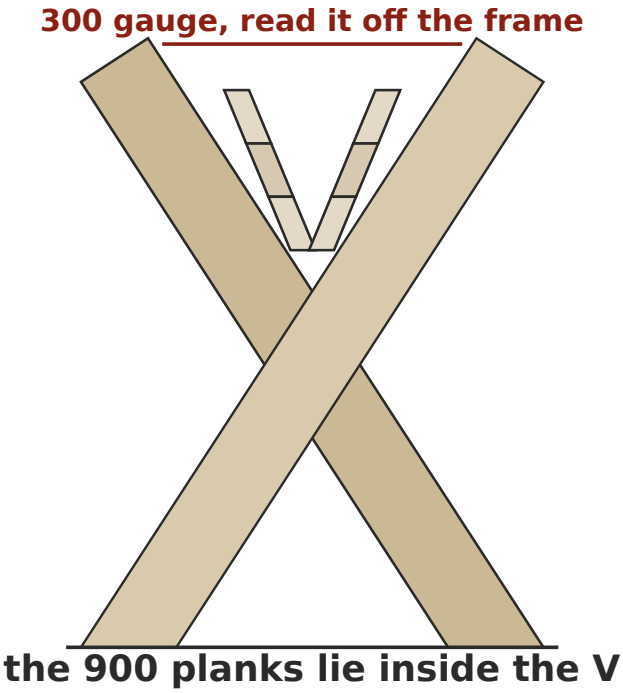
- **Top = 300, inside corner to inside corner.** Read it off the standing frame - it is the check, never a cut-to. Every X the same.
- **Feet cut at 33°** so they sit flat. Method: clamp a plank, leg flush, one centre screw into the middle of the plank.
- **The two legs never notch.** They lie parallel, one thickness apart, on the same centre line. Both feet therefore reach the floor. That gives an inside plank and an outside plank.
- **Spacer blocks at the shoulder.** Small blocks fill the gap between the angled leg head and the underside of the first planks. Drawn from the ~60 and ~75 mm marks in your shots.
- **One wall flush, one proud.** The back planks run **flush to the outside leg**; the front planks stand **one plank thickness proud** (23 mm for 23 mm stock), as in your shots.

EVERY CUT, DRAWN



Two angles hold the whole thing. **33°** on the leg. **22.5°** in the V. The rest square.

Where the planks sit end view



End view: the planks sit inside the V.

- Not a tray on top. It drops down inside the X.
- Both sides lean in at **22.5°** to a narrow bottom.
- Leg tops stand proud above the rim.
- Lowest plank: bottom edge cut at 22.5°, so it sits level.
- Next plank lands one thickness higher. **Trim one thickness off its long edge.** Tops come level.

The two angles that fool everyone

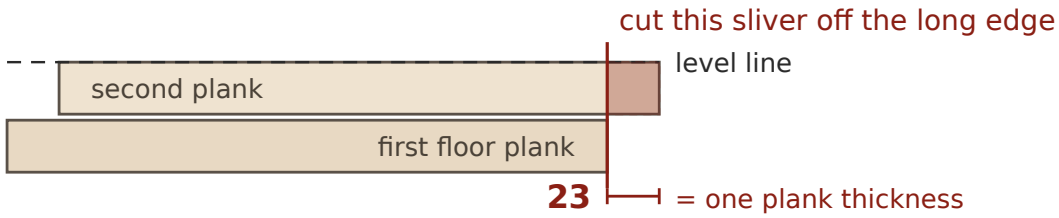
- 33° is the **leg** - the foot cut and the lean, one saw setting. Nothing else is 33°.
- 22.5° is the **V** - how far each side leans in, and the cut on the lowest plank's bottom edge to match it. A chosen call: it beds the plank into the V, it is **not** derived from the 33°.
- Each side above that is **scribed** off an offcut stood in the V, not set to a number.

The trim, up close

the one lengthwise cut

THE ONLY LENGTHWISE TRIM

23 is one plank thickness. 19 stock: trim 19.



How to get it level

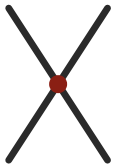
- First floor plank beds into the V on its 22.5° edge.
- Second plank lands **one thickness higher**. That is the gap.
- Trim **one plank thickness** off its long edge. The tops then sit level.
- 23 is your stock. 19 stock: trim 19. The thickness is the rule, not the number.
- Trim it before you screw. Screwed down and you are planing a fixed course.

The build in four stages

picture sequence

Four stages, in order. Do not start the sides until the walls are screwed down and level.

1

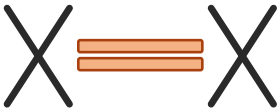


33° feet · screw the centre

The X legs.

- Cut one end of each 700 leg at 33°.
- Upper leg on top of the lower. Centre 350, 40 across.
- One screw through the middle. Make two, mirrored.
- Both sets at the same 300 top gauge.

2



first two in the V

Stand them, start the floor.

- Two X sets parallel, set to the 300 gauge.
- First 900: cut one long edge at **22.5°**, inner edge.
- Second 900 lands one thickness higher.
- Trim one thickness off its long edge.

3



4 front + 4 back

The rest of the 900s.

- 4 planks front, 4 back.
- Flush with the left and right X.
- Back sits one thickness proud.
- Screw down front and back.

4



sides off an offcut

The sides.

- Hold an offcut in the V. Mark the angle.
- Cut the first side piece. Fit it.
- Next two: mark off the one below. Cut, glue, screw on top.
- Mirror the other end.

Stage by stage 1 – 2

#	STAGE	WHY
1	**The X legs.** • Four legs, 700 long. Cut **one end of each** at 33° - all four off the **same saw setting**. • Find the centre of each leg: 350 along, 40 across (half of 80). • Clamp a straight plank to the table. Press the 33° end flush against it. • Lay the second leg alongside the first, **parallel, one thickness to the side**, to make the X. Pilot hole at the centre. • Drive a 35 mm part-threaded screw through the near plank so it bites the far one — one thickness of travel. • Then 3-4 more screws, or swap to a 6 mm or 10 mm bolt through the crossing. • Set the standing frame so the **top gauges 300, inside corner to inside corner** - read it, don't cut to it. Make the second set the same.	• One saw setting for all four feet = feet that sit level. Cut them one at a time and they fight. • The clamped plank is the reference - it stops the X leaning while you screw. • The legs do not notch: parallel, one thickness apart, both feet down. That gives an inside leg and an outside leg. • A single screw lets the X scissor. Extra screws or a bolt lock the crossing. • 300 top gauge on both sets = the box sits level and the walls line up.
2	**Stand them and lay the first two planks.** • Stand the two X sets parallel, on the ground. • Lay the first two 900 planks in the V, one front, one back. • Cut **one long edge of the first plank at 22.5°** - the inner edge where it meets the far leg - so it beds into the V. • The second plank lands **one plank thickness** higher, so **trim one plank thickness off** its long edge - 23 mm for 23 mm stock - and both tops come level.	• Both X sets parallel = the box goes up straight. • The 22.5° on the inner edge is what lets the plank bed into the V instead of rocking on the leg corner. • Level first planks = level box. This is the one that sets it.

Stage by stage 3 – 4

#	STAGE	WHY
3	**Lay the rest of the 900s.** • Carry on stacking: 4 planks on the front, 4 on the back, one on top of the other. • All flush with the left and right X planks. • The back stacks one plank thickness proud (23 mm here), behind the outside leg. • Both sides finish at the same height. • Screw the stack down, front and back, into the X frames.	• Flush to the X = straight walls, no stepping. • The back offset clears the leg, so both walls meet the frame the same way. • Equal height both sides = a level rim.
4	**The sides, scribed off an offcut.** • Hold an offcut horizontal in the bottom of the V. Draw a line from inside to mark the angle of the first side piece. • Cut to the line, fit it in the V. • Hold a longer offcut above the first, mark off it, cut the second, glue and screw it on top of the first. • Third the same, on top of the second. • Do the same for the opposite end.	• The V is not a set angle you can measure off a drawing - you scribe each piece off the one below. • Glue **and** screw: end grain is weak, the glue takes the load, the screws hold it while it sets. • Three pieces fill the end from the floor to the rim.